

Configuration

💡 Only bearings 3 & 4 are shown (these are the only bearings that failed in the dataset)

Select Bearing

3

Select Axis

y



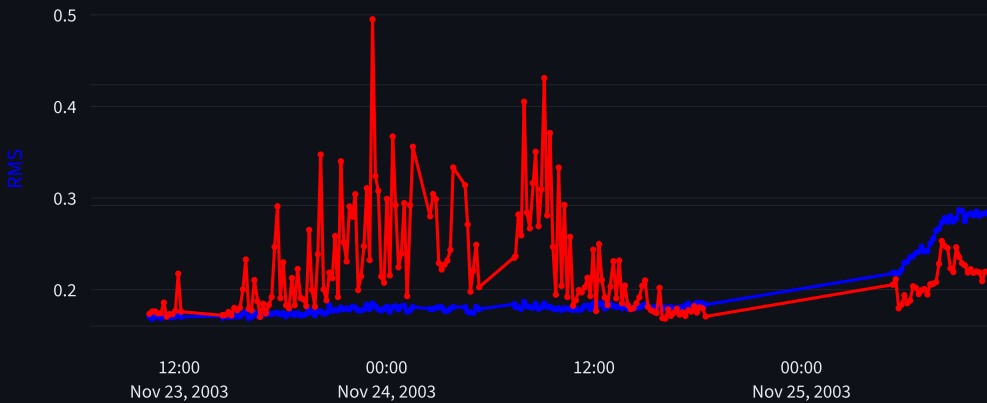
Bearing Failure Prediction Dashboard

Real-time monitoring and RUL prediction | 2.88h MAE (critical zone)

Current RMS	Current Kurtosis	Actual RUL	Predicted RUL
0.4952	4.06	0.0 hours	1.3 hours
			↑ 1.3h error

🚨 **CRITICAL:** Schedule emergency maintenance immediately!

Degradation Trends (Last 200 Measurements)

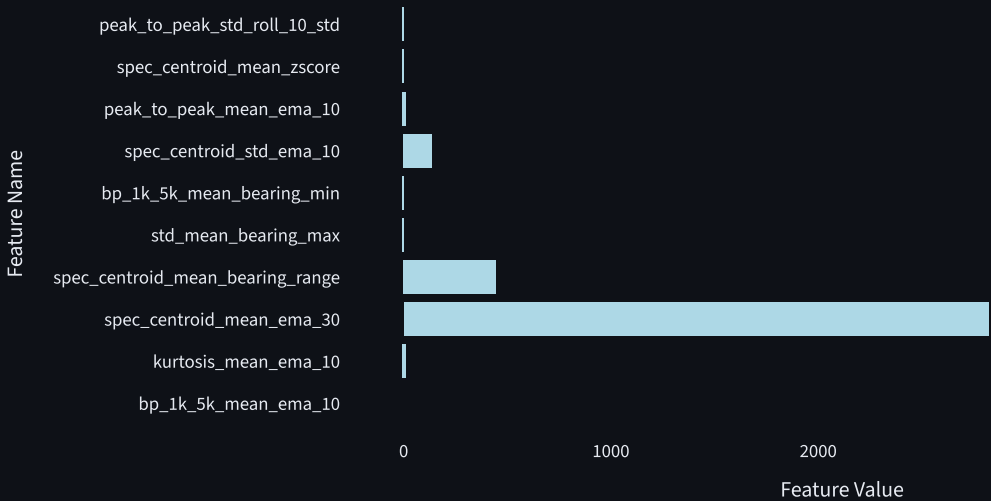


RUL Degradation Over Time



Top 10 Most Important Features (Current Values)

Feature Values at Current Time (Bearing 3-y)



Summary Statistics

Metric	Value	Metric	Value
Mean RMS	0.1972	Mean Kurtosis	2.44
Max RMS	0.4952	Max Kurtosis	12.71
Min RMS	0.1683	Min Kurtosis	0.31
Current RMS	0.4952	Current Kurtosis	4.06

Dataset Information

Total Measurements	Initial RUL	Time to Failure
200	55.9 hours	2.3 days

Model Details:

- **Algorithm:** LightGBM v2 (Optuna-tuned, 80 trials)
- **Critical Zone MAE:** 2.88 hours (0-50h RUL range)
- **Overall MAE:** 13.42 hours
- **R²:** 0.9852 (98.5% variance explained)
- **Dataset:** NASA IMS Bearing Dataset
- **Features:** 50 selected from 380 engineered features
- **Author:** Anudeep | Mechanical Engineer → Data Scientist